

Quantum Computing: Origin & Global Impact

A High-Level Overview of How It Started, Why It Was Built, and How It Changes the World

1. What is Quantum Computing & Why Was It Invented?

For decades, standard silicon computers have grown faster and smaller. However, when trying to simulate nature—such as complex chemical reactions, weather systems, or atomic behavior—traditional computers reach a wall. They struggle because nature follows the laws of physics at an atomic scale, which do not behave like standard machine code.

Quantum computing was proposed to bridge this gap: building a computer that naturally mimics the fundamental laws of nature to solve calculations that would take traditional supercomputers thousands of years to complete.

2. How It Was Founded: The Historical Journey

The field originated through key breakthroughs by physicists and computer scientists trying to simulate physics:

HISTORICAL MILESTONES OF QUANTUM COMPUTING

1981 — Richard Feynman's Proposal

Nobel laureate Richard Feynman famously pointed out that classical computers cannot efficiently simulate quantum physics, suggesting that we build a computer based on quantum mechanics instead.



1985 — David Deutsch & The Universal Quantum Computer

Physicist David Deutsch described the theoretical model for a universal quantum computer, proving it could perform algorithmically distinct computations from standard computers.



1994 — Peter Shor's Encryption Breakthrough

Peter Shor developed an algorithm capable of factoring large numbers exponentially faster than classical computers, demonstrating real-world disruption for global cybersecurity.



2010s to Present — Practical Hardware & Quantum Supremacy

Major tech companies and research institutes transitioned quantum computing from theoretical physics into physical devices operating at sub-zero temperatures.

3. The Usefulness of Quantum Computing

Quantum computing is not intended to replace personal laptops or smartphones. Instead, its immense usefulness lies in solving complex computational bottlenecks across major industries:



Medicine & Drug Discovery

Simulates molecular and protein interactions at an atomic level, allowing researchers to discover life-saving drugs and vaccines in days rather than decades.



Global Supply Chains & Logistics

Instantly calculates optimal delivery routes and schedules across millions of variables, drastically cutting global fuel consumption and supply chain bottlenecks.



Climate Science & Clean Energy

Helps engineer highly efficient battery technologies and discovers new catalysts to capture carbon dioxide directly from the atmosphere.



Financial Risk & Economics

Analyzes vast market datasets to optimize portfolio investments, detect fraud in real-time, and model global economic changes.

4. Summary Comparison: Everyday vs. Quantum Impact

Aspect	Standard Classical Computers	Quantum Computers
Primary Purpose	General tasks (browsing, docs, apps, videos)	Simulating nature & massive complex systems
Core Advantage	Reliable sequential processing for daily life	Solves impossibly large multi-variable problems
Key Impact Area	Personal productivity & global software	Scientific breakthroughs, healthcare, and materials